REPORT 013: EPOCH 013 — PRESET LIBRARIAN, 14 FACTORY PRESETS, LEGACY/JSON PARSER & A/B DIFF/MERGE ENGINE

2026-11-04 01:22:12 | Implementation of src/config_parser.rs and src/librarian.rs with 14 factory presets, JSON and mappings/default.txt import/export, 5 Quick Access slots, and A/B Preset Diff & Selective Merge

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Prompt

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Epoch 13 of 20: Implement the complete Preset Librarian in `src/librarian.rs` and
↪  `src/config_parser.rs`: (1) include all 10 original factory presets from
↪  `nkEditor3` (`Default`, `Synth Lead`, `Drums`, `DAW Transport`, `DJ Mixer`, `Nord
↪  Drum 2` for `nanoKONTROL2`; `Default MidiMix`, `MidiMix Synth`, `MidiMix DJ`,
↪  `MidiMix Nord Drum 2` for `Akai MIDImix`) PLUS 4 new factory presets (`8opPerc
↪  nanoKONTROL2 Live Rig`, `8opPerc MIDImix Compatibility`, `Konduktiva 8-Lane Macro
↪  Surface`, `Dual-Controller Split Rig`); (2) support importing and exporting both
↪  JSON preset files and legacy `mappings/default.txt` format; (3) provide 5 Quick
↪  Access preset slots, search, tag/group filtering, and favorites; and (4)
↪  implement an A/B Preset Diff & Selective Merge view in `egui` comparing any two
↪  presets across mappings, channels, ranges, and control values. Add automated
↪  tests (with `CARGO_BUILD_JOBS=2`), git commit all changes, and compile
↪  `reports/report-013.html` from `reports/report-013.md` via
↪  `/home/renick/claude/html-report-generator/build-report.sh --pdf` and commit.
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TL;DR

Implemented the complete Preset Librarian in src/librarian.rs and legacy/JSON configuration parser in src/config_parser.rs: (1) all 10 original nkEditor3 factory presets plus 4 new nkEditor4/8opPerc factory presets (14 total); (2) lossless import and export for both JSON preset files and legacy mappings/default.txt text configurations (including [device akai-midimix] and note:N syntax); (3) 5 Quick Access preset slots, search, group/tag/device filtering, and favorites; and (4) an interactive A/B Preset Diff & Selective Merge engine and egui view. Verified with 50 passing Rust tests and 10 passing Node.js tests (60 total).

Plain-Language Overview: What Changed and Why It Matters

When performing live or working in a studio with multiple synthesizers, drum machines, and software environments, a musician needs to switch the entire layout of their MIDI controller in a single click—without losing track of custom tweaks or breaking old mapping files saved from previous versions of the software. Furthermore, when someone creates two variations of a setup (for example, a standard mapping and a specialized `Nord Drum 2` setup), they often want to see *exactly* which knobs or buttons changed between the two presets and copy only a few specific controls from one preset into another.

In Epoch 13, we built `src/config_parser.rs` and `src/librarian.rs` to solve four practical problems:

- **14 Built-in Factory Presets (`create_all_factory_presets`)**: We preserved all **10 original factory presets** from `nkEditor3` (6 for the Korg `nanoKONTROL2` and 4 for the Akai `MIDI Mix`) so existing workflows behave identically, and added **4 new factory presets** tailored for `nkEditor4`,

8opPerc, and Konduktiva live performance (8opPerc nanoKONTROL2 Live Rig, 8opPerc MIDImix Compatibility, Konduktiva 8-Lane Macro Surface, and Dual-Controller Split Rig).

- **Lossless Legacy Text (mappings/default.txt) & JSON Import/Export (src/config_parser.rs)**: Users can load and save full Preset JSON files (which store metadata, tags, continuous target values, button states, and secondary controller mappings) as well as human-readable mappings/default.txt text files (supporting [device akai-midimix], [track1]..[track8], [transport], [master], and note:N MIDI Note syntax).
- **5 Quick Access Slots, Instant Search, Group/Tag Filters & Favorites (PresetLibrarian)**: The Librarian header provides 5 one-click Quick Access preset buttons (quick_access_slots), a real-time search box matching preset names, descriptions, groups, and tags, category filters, and favorite star toggles (toggle_favorite).
- **A/B Preset Diff & Selective Merge View (diff_presets, merge_selected_controls, render_preset_librarian_panel)**: Users can pick any two presets (Preset A and Preset B) to inspect a side-by-side table of every control that differs across MIDI CC/Note numbers, channels, min/max ranges, response curves, button behaviors, labels, and saved control values—and selectively merge checked controls into a new hybrid preset.

1. Complete Catalog of All 14 Factory Presets

The table below maps all 14 built-in factory presets in src/librarian.rs to their target hardware controller, group, and mapping architecture:

#	Preset Name	Preset ID	Target Device	Group	Key Mapping Details & Rust Factory Function
1	Default	factory-default	nanokontrol2	Factory	51-control 1:1 CC passthrough on Ch 1 (factory_preset_default_nk2)
2	Synth Lead	factory-synth-lead	nanokontrol2	Factory	Knobs -> Filter/Res/Env/FX CCs 74, 71, 73, 72, 91, 93, 1, 11; Sliders -> OSC/Master CCs 16..22, 7 (factory_preset_synth_lead_nk2)

#	Preset Name	Preset ID	Target Device	Group	Key Mapping Details & Rust Factory Function
3	Drums	factory-drums	nanokontrol2	Factory	Knobs -> Drum Tune/Decay CCs 10, 42..48; Sliders -> Drum Levels; Solo -> Momentary Trigger (factory_preset_drums_nk2)
4	DAW Transport	factory-daw-transport	nanokontrol2	Factory	Mackie-compatible transport CCs (Play=94, Stop=93, Rec=95, Rew=91, Fwd=92, Loop=86) + 8 mixer strips (factory_preset_daw_transport_nk2)
5	DJ Mixer	factory-dj-mixer	nanokontrol2	Factory	Deck A (Tracks 1-4) & Deck B (Tracks 5-8) 3-band EQ + Filter CCs 22..29, X-Fade, Cue, Play, Sync (factory_preset_dj_mixer_nk2)
6	Nord Drum 2	factory-nord-drum-2-nk2	nanokontrol2	Nord Drum 2	Tracks 1-6 -> Pads 1-6 (Ch 1-6, Knob=SPECTRA CC 30, Slider=DECAY CC 50); Tracks 7-8 -> Global FX on Ch 10 (factory_preset_nord_drum_2_nk2)

#	Preset Name	Preset ID	Target Device	Group	Key Mapping Details & Rust Factory Function
7	Default MidiMix	factory-default-midimix	akai-midimix	MidiMix Factory	60-control 1:1 identity mapping with CC knobs/faders and Note buttons (factory_preset_default_midimix)
8	MidiMix Synth	factory-midimix-synth	akai-midimix	MidiMix Factory	Row 1 Knobs -> Synth CCs 74, 71, 73, 72, 91, 93, 1, 11; Faders -> OSC/Env/FX/Master levels (factory_preset_midimix_synth)
9	MidiMix DJ	factory-midimix-dj	akai-midimix	MidiMix Factory	Row 1 Knobs -> EQ High CCs 22..29; Faders -> Ch 1-8 Vol; Mute -> Cue; Solo -> PFL (factory_preset_midimix_dj)
10	MidiMix Nord Drum 2	factory-midimix-nord-drum-2	akai-midimix	Nord Drum 2	32 continuous controls: Row1=SPECTRA, Row2=FILTER_FREQ, Row3=TIMBRE, Fader=LEVEL across Pads 1-6 (Ch 1-6) + Global FX (Ch 10) (factory_preset_midimix_nord_drum_2)

#	Preset Name	Preset ID	Target Device	Group	Key Mapping Details & Rust Factory Function
11	`8opPerc nanoKONTROL2 Live Rig`	`factory-8opperc-nk2-live`	`nanokontrol2`	`8opPerc`	Native 8-lane percussion rig (`kick..shaker` Density, Gain/Macro, Solo, Mute, Reroll, Form Prev/Next, Timbre Set, Snapshot) (`factory_preset_op8_nanokontrol2_live_rig`)
12	`8opPerc MIDImix Compatibility`	`factory-8opperc-midimix-compat`	`nanokontrol2`	`8opPerc`	Cross-message `CC -> Note` and `CC -> CC` emulation translating `nanoKONTROL2` into Akai `MIDI Mix` hardware events (`factory_preset_op8_midimix_compatibility`)
13	`Konduktiva 8-Lane Macro Surface`	`factory-konduktiva-macro-surface`	`nanokontrol2`	`8opPerc`	Ch 2 expressive surface with `SCurve` macro knobs (`CC 102..109`, range `8..120`) and `Logarithmic` stem faders (`CC 110..117`) (`factory_preset_konduktiva_macro_surface`)

#	Preset Name	Preset ID	Target Device	Group	Key Mapping Details & Rust Factory Function
14	Dual-Controller Split Rig	factory-dual-controller-split-rig	nanokontrol2	8opPerc	Dual-controller preset pairing nanoKONTROL2 (8opPerc Live Rig on Ch 1) + secondary akai-midimix (Nord Drum 2 32-param synth control on Ch 1-6 & Ch 10) (factory_preset_dual_controller_split_rig)

2. Legacy Text (‘mappings/default.txt‘) & JSON Parser (‘src/config_parser.rs‘)

The table below connects the human-readable configuration features to their implementation in src/config_parser.rs:

Feature	Plain-Language Role	Rust Symbol in ‘src/config_parser.rs‘
Legacy .txt Parser	Parses [device], [track1]..[track8], [transport], [master], note:N syntax, ch1..ch16, toggle/momentary, and trailing "label" strings with line-numbered errors	parse_legacy_mapping_text & ConfigParseError
Legacy .txt Formatter	Formats any DeviceMappingConfig back into clean, human-readable mappings/default.txt syntax for export	format_mapping_to_text
JSON Preset Import/Export	Serializes and deserializes full Preset structs (including metadata, tags, continuous targets, button states, and secondary_mapping)	parse_preset_json & format_preset_json
Format Auto-Detection	Automatically detects whether an imported file is JSON ({...}) or legacy text ([track1]...)	import_preset_auto

3. A/B Preset Diff & Selective Merge Engine (‘src/librarian.rs‘)

- **Field-Level Comparison (diff_presets, PresetDiffReport, ControlDiffEntry, FieldDiff)**: Compares any two presets (preset_a vs preset_b) across every control path (track1.knob, transport.play, etc.) and records differences in mapping (CC/Note type and number), channel (Ch 1..16), range (min_value..max_value), curve (Linear, Exponential, Logarithmic, SCurve), invert, behavior (Toggle vs Momentary), on_off values, label, and saved target_value / button_state.

- **Selective Per-Control Merge (`merge_selected_controls`, `PresetLlibrarian::execute_selective_merge`)**: Allows the user to check or uncheck individual control paths in `selected_merge_paths` and merge only those selected controls from Preset B onto Preset A, producing a conflict-free user preset.
- **Interactive `egui` Panel (`render_preset_librarian_panel`)**: Renders the 5 Quick Access slot buttons, search/favorite/group filter bar, scrollable preset browser with favorite toggle, `Set A` / `Set B` diff selectors, `Dup` / `Del` actions, and the interactive A/B Diff & Selective Merge checklist.

Tests

QA

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**Total Tests**: `60` passing (`50` Rust tests + `10` Node.js tests), `0` failing, `0` compiler warnings.

**Safety & Resource Discipline**: All builds and tests executed with `CARGO_BUILD_JOBS=2` in headless mode
without launching blocking GUI windows or live audio playback.

**Verdict**: **PASS**.
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Files touched

- ‘src/config_parser.rs‘ (Created — 456 lines): Implemented ‘ConfigParseError‘, ‘parse_legacy_mapping_text‘, ‘format_mapping_to_text‘, ‘parse_preset_json‘, ‘format_preset_json‘, and ‘import_preset_auto‘.
- ‘src/librarian.rs‘ (Created — 1,073 lines): Implemented ‘PresetMetadata‘, ‘Preset‘, all 14 factory preset builders (‘create_all_factory_presets‘), ‘FieldDiff‘, ‘ControlDiffEntry‘, ‘PresetDiffReport‘, ‘diff_presets‘, ‘merge_selected_controls‘, ‘PresetLibrarian‘, ‘render_preset_librarian_panel‘, and 5 comprehensive unit and headless ‘egui‘ tests.
- ‘src/lib.rs‘ (Modified): Exported ‘pub mod config_parser;‘ and ‘pub mod librarian;‘.
- ‘reports/report-013.md‘, ‘reports/report-013.html‘, ‘reports/report-013.pdf‘ (Created): Epoch 013 report in Markdown, grayscale HTML, and PDF formats.

Test coverage (60 total)

QA verdict

PASS

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